MGT 207: Operations Management

Process Analysis & Capacity Management — The Car Wash Simulation Lab

PHASE 1 (20 min)
Setup & Briefing
- AutoSpa P&L rules
- 5 Dashboard metrics

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PHASE 2 (20m + 10m)
Round 1: "Cold Run"
- Run Challenge 1 ONLY
- **STOP** before Challenge 2

→

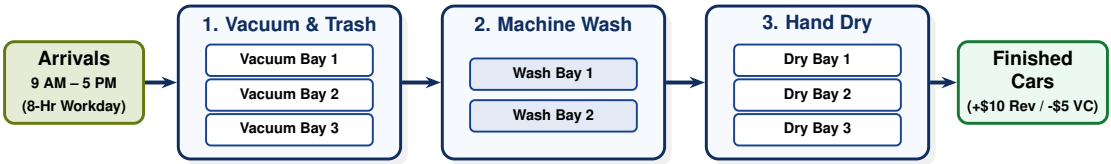
PHASE 3 (55 min)
Physics of Capacity
- Little's Law & Subway
- Reverse-engineering Ch. 1

→

PHASE 4 (55 min)
Graded Challenge 2
- Run Challenge 2 (Graded)
- Executive debrief

"In operations, intuition gets you started—physics gets you paid. Today you will manage the exact same facility twice: once by gut feel, and once by operational science."

Your Role: Operations Manager at AutoSpa Car Wash



Rule 1: Spend 100% of $3,000,000 Budget

- Fixed **$3.0M Capex budget** to upgrade station speeds.
- Must spend 100% every run (no leftover cash):

$$(\text{Vacuum \$M}) + (\text{Wash \$M}) + (\text{Dry \$M}) = \$\mathbf{3.0}\text{M}$$

Rule 2: Maximize Daily Operating Profit

- Equipment depreciates over multi-year horizons.
- $3M Capex **does NOT hit today's daily P&L**.
- Sole goal: maximize **Daily Operating Profit**.

The Daily P&L & The Two Simulation Challenges

Daily Operating P&L (8-Hour Day)

- **Hours:** 9:00 AM – 5:00 PM (8 hours/day).
- **Revenue:** $10.00 / car.
- **Variable Cost:** $5.00 / car *(constant)*.
- **Margin:** $10 − $5 = **$5.00** / car.
- **Fixed Cost:** **$675** / day *(break-even* = 135 *cars)*.

Challenge 1: No Variation (Steady State)

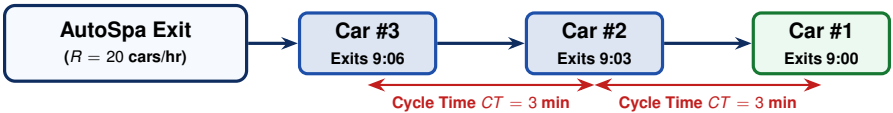
- **Uniform Arrivals:** 37.5 cars/hr all day; fixed bay speeds.
- **Gatekeeper:** Must hit max profit to unlock Challenge 2!

Challenge 2: Variation (GRADED in Round 2!)

- **Real World:** Noon rush (**12–1 PM**) + random hourly speeds.
- **STOP Rule:** Do **NOT** start Challenge 2 in Round 1!

FORMULA REMINDER: Daily Profit = ($5 × Cars Washed) − $675 | Cars Washed per Day = 8 Hours × Hourly Throughput Rate (R)

Dashboard Vocabulary (Part 1: Speed & Pacing)



1. Throughput Rate (R) — Flow Rate

- **Meaning:** Rate at which finished cars exit.
- **Units: Cars / hour** (e.g., $R = 20$ cars/hr).
- **Impact:** Drives daily output (8 hrs $\times$ R).

2. Cycle Time (CT) — Exit Pacing

- **Meaning:** Time **between consecutive cars** exiting.
- **Units: Minutes / car** (inverse of R).
- **Example:** At 20 cars/hr, 1 car exits every $60/20 = \mathbf{3}$ min.

FORMULA REMINDER: Cycle Time (CT in min/car) $= \frac{60 \text{ minutes/hour}}{\text{Throughput Rate } (R \text{ in cars/hour})}$ | Throughput Rate (R) $= \frac{60 \text{ minutes/hour}}{\text{Cycle Time } (CT)}$

Dashboard Vocabulary (Part 2: Congestion & Utilization)

3. Throughput Time (T)

- **Meaning:** Total time **one car** spends inside from entry to exit.
- Includes **queue wait** + **bay time**.
- **Units:** Minutes (e.g., 15 min).

4. Work-in-Process (WIP)

- **Meaning:** Total cars inside the facility at any moment.
- Includes **queues** + **bays**.
- **Units:** Cars (e.g., 5 cars).

5. Capacity Utilization (U)

- **Meaning:** Share of max capacity actually used:

$$U = \frac{R}{\text{Capacity}} \times 100\%$$

KEY DISTINCTION: **Cycle Time (CT)** = time between exiting cars (3 min);
Throughput Time (T) = total time *your* car spends inside (15 min)!

FORMULA REMINDER: Utilization (U) $= \frac{R}{\text{Capacity}} \times 100\%$ | WIP $= R$ (cars/hr) $\times$ T (hours) $= \frac{T \text{ (minutes)}}{CT \text{ (minutes/car)}}$

Quick Check: Reading the Dashboard (2-Minute Warm-Up)

Scenario

A single-stage Car Wash runs at **Throughput Rate** $R = 20$ **cars/hour** (max Capacity $= 25$ cars/hr, so Utilization $U = 80\%$). Each car spends **Throughput Time** $T = 15$ **minutes (**0.25 **hours)** inside the facility.

Question 1: Cycle Time (CT)

How many **minutes elapse between** consecutive cars exiting the bay?

Compute CT (minutes / car)

Question 2: System WIP (Little's Law)

If 20 cars/hr enter and each stays 0.25 hr (15 min), how many **cars are inside** at any moment?

Compute WIP (cars in system)

FORMULA REMINDER: $CT = \frac{60 \text{ min/hr}}{R \text{ (cars/hr)}}$ | $\text{WIP} = R \text{ (cars/hr)} \times T \text{ (hr)} = \frac{T \text{ (min)}}{CT \text{ (min/car)}}$ | $U = \frac{R}{\text{Capacity}} \times 100\%$

LAB SESSION 1 — Run Challenge 1 and STOP

RUN CHALLENGE 1 AND STOP!

Do NOT start Challenge 2 yet — Challenge 2 will be GRADED in Round 2!

Your Mission (Challenge 1 Only)

1. **Run Challenge 1 (No Variation):** Spend your $3M across Vacuum, Wash, and Dry to hit **$825 max profit**.
2. **STOP Once Solved:** Do **not** click Challenge 2!

Record 2 Things Before Break

1. Your winning **(Vacuum $M, Wash $M, Dry $M)** split and number of trials needed.
2. Your winning **Cars Washed**, **Cycle Time** (CT), and **Throughput Time** (T).

TIMING: 20-Min Challenge 1 Run → 10-Min Coffee Break → Reconvene for Debrief & Challenge 2!

Round 1 Debrief: Little's Law (Part 1: The Fruit Bowl Intuition)

Little's Law (Fundamental Law of Process Flow):

$$\textbf{WIP} = R \times T \iff T = \frac{\textbf{WIP}}{R} = \textbf{WIP} \times CT$$

Intuition: The Kitchen Fruit Bowl

- Bowl holds **WIP** $= 10$ **apples**.
- You eat $R = 1$ **apple/day** (and replace 1/day).
- Average time an apple sits in the bowl:

$$T = \frac{\text{WIP}}{R} = \frac{10 \text{ apples}}{1 \text{ apple/day}} = \textbf{10 days!}$$

Unit Consistency Rule

- Match time units for R and T:
- **In Hours:**

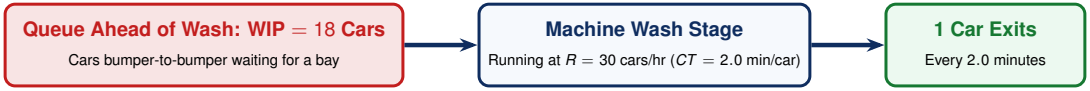
$$\text{WIP} = R \text{ (cars/hr)} \times T \text{ (hours)}$$

- **In Minutes:**

$$T \text{ (min)} = \text{WIP} \times CT \text{ (min/car)}$$

FORMULA REMINDER: $\text{WIP} = R \text{ (cars/hr)} \times T \text{ (hr)}$ | $T \text{ (min)} = \text{WIP} \times CT \text{ (min/car)}$ | $CT = \frac{60 \text{ min/hr}}{R \text{ (cars/hr)}}$

Little's Law (Part 2: Why Queues Explode Customer Wait Time)



Why Wait Time (T) Spikes

- 18 **cars** wait ahead of Wash ($R = 30$ cars/hr).
- Wash finishes 1 car every $CT = 60/30 = \mathbf{2.0}$ min.
- Extra wait for the 18th car:

$$18 \text{ cars} \times 2.0 \text{ min/car} = \mathbf{36.0} \text{ extra min!}$$

Managerial Takeaway

- At any speed R, **Queue WIP and Wait Time (T) move in lockstep**.
- To cut customer wait time (T), you must **prevent WIP queues from forming**!

FORMULA REMINDER: Queue Wait Time (min) = (Cars Waiting in Queue) $\times$ Cycle Time (CT) $\quad | \quad T = \frac{\text{WIP}}{R} = \text{WIP} \times CT$

The Physics of Flow: Identifying the System Bottleneck

What Is a System Bottleneck?

The stage with the **lowest hourly capacity** (or **longest cycle time**):

System Capacity $=$
Lowest Stage Capacity (Slowest Stage)

Throughput (R) $=$
Lower of (**Demand**, **System Capacity**)

- **Weakest Link Rule:** Faster stages cannot push more cars out the exit than the slowest stage allows.

Two Ways to Spot the Bottleneck

1. **On Paper (Calculation):**
   - Compute cars/hr for each stage; the **lowest number** is the bottleneck.
2. **On the Screen (Visual Queue):**
   - Look for the **biggest WIP queue right in front of a station** ($\sim$ 100% utilization), while downstream bays sit starved!

FORMULA REMINDER: Stage Capacity = (Number of Parallel Bays) $\times$ (Capacity of 1 Bay) | System Capacity = Lowest Stage Capacity

Mini-Case Part 1: Single-Line Capacity Analysis (Subway)

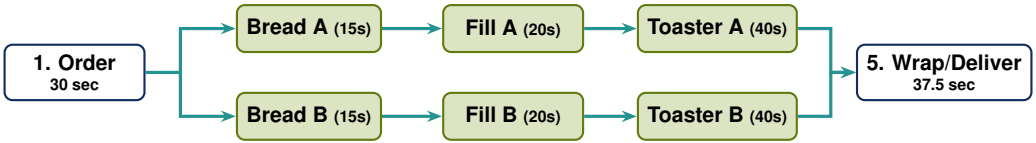


Pair Exercise (2 Minutes — 2 Quick Questions)

1. **Bottleneck & System Capacity:** Which station is the slowest, and what is the maximum hourly output (subs/hr)?
2. **Rush-Free Throughput Time (T):** How many seconds does 1 sandwich take from start to finish with zero queue waiting?

FORMULA REMINDER: Stage Capacity $= \frac{3{,}600 \text{ sec/hr}}{\text{Process Time (sec)}}$ | System Capacity = Lowest Stage Capacity | Rush-Free T = Sum of All 5 Stage Times

Mini-Case Part 2: Parallel Capacity & The Shifting Bottleneck Trap

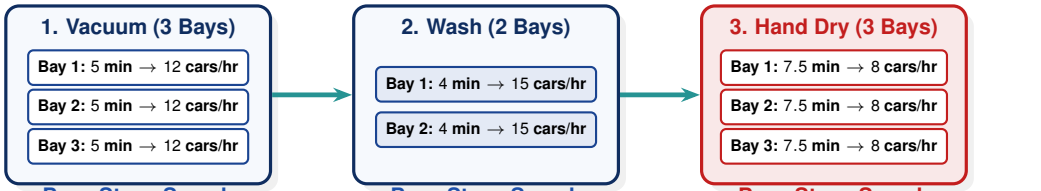


Pair Exercise (2 Minutes — Manager Adds a 2nd Parallel Bread → Fill → Toaster Line)

1. **New Bottleneck:** With 2 parallel lines for Bread, Fill, and Toaster, which station is now the slowest?
2. **New System Capacity:** What is the new maximum hourly output (subs/hr)—did doubling the Toaster double output?

FORMULA REMINDER: Parallel $CT = \frac{\text{Single-Line Time}}{\text{Number of Parallel Lines}}$ | Stage Capacity $= (\text{Parallel Lines}) \times \frac{3{,}600}{\text{Process Time}}$ | System Capacity $=$ Lowest Stage Capacity

From Sandwiches to Cars: The Exact Same Parallel Setup!



Parallel Bays Working Side by Side

Vacuum: 3 bays × 12 = **36** cars/hr | **Wash:** 2 bays × 15 = **30** cars/hr | **Hand Dry:** 3 bays × 8 = **24** cars/hr

FORMULA REMINDER: Stage Capacity = (Number of Parallel Bays) × (Cars/Hr per Bay) | Base Speeds ($0 Spent): Vacuum 36/hr, Wash 30/hr, Dry 24/hr

At $0 Spent, What Happens on Day 1?

1. Base Speeds ($0 Spent)

- **Vacuum (3 Bays): 36.0** cars/hr
- **Machine Wash (2 Bays): 30.0** cars/hr
- **Hand Dry (3 Bays): 24.0** cars/hr **(BOTTLENECK!)**

Result: Hand Dry caps the entire car wash at **24.0** cars/hr.

2. Baseline Output & Profit ($0 Spent)

- **Cars Washed (8 Hours):**

$$8 \text{ hrs} \times 24.0 \text{ cars/hr} = \mathbf{192} \text{ cars/day}$$

- **Baseline Daily Profit:**

$$(\$5 \times 192) - \$675 = \$\mathbf{285}/\mathbf{day}$$

- **Goal:** Use $3.0M Capex to lift speed from 24.0 $\rightarrow$ **37.5** cars/hr!

FORMULA REMINDER: System Capacity = Lowest Stage Capacity (24 cars/hr) | Cars/Day $= 8 \times R$ | Daily Profit = ($5 $\times$ Cars/Day) $-$ $675

Step 1: Reading the Upgrade Price Tags on the Screen

What Does $1 Million Buy at Each Stage? (Bottom of Simulation Screen)

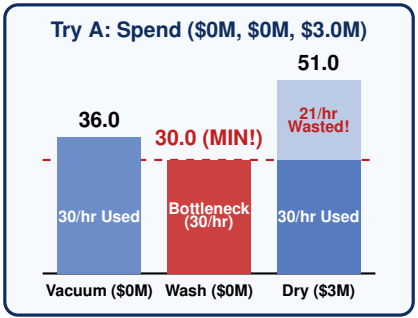
Each station displays its **Base Speed ($0)** and **Speed Gained per $1M Spent**:

STAGE	BAYS	BASE SPEED ($0)	GAIN PER $1M	NEW CAPACITY
1. Vacuum & Trash	3 Bays	36.0 cars/hr (12.0/hr per bay)	+**1.5** cars/hr (+0.5/hr per bay)	$36.0 + 1.5 \times$ ($M Spent)
2. Machine Wash	2 Bays	30.0 cars/hr (15.0/hr per bay)	+**15.0** cars/hr (+7.5/hr per bay)	$30.0 + 15.0 \times$ ($M Spent)
3. Hand Dry	3 Bays	**24.0** cars/hr (8.0/hr per bay)	+**9.0** cars/hr (+3.0/hr per bay)	$24.0 + 9.0 \times$ ($M Spent)

QUESTION: Hand Dry is slowest (24/hr). What happens if we dump **all $3M into Dry**, or split it evenly (**$1M** / **$1M** / **$1M**)?

FORMULA REMINDER: New Stage Capacity = Base Speed + (Speed Gained per $1M) $\times$ ($M Spent at Stage)

Step 2A: What If We Dump All $3M into the Bottleneck (Hand Dry)?



Naive Try A: ($0M Vacuum, $0M Wash, $3.0M Dry)

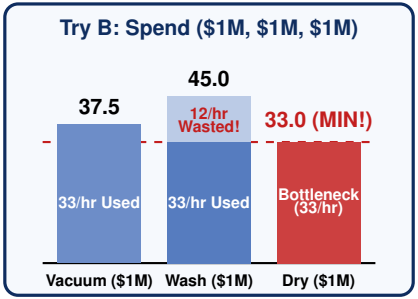
- **Vacuum ($0M):** 36.0 cars/hr
- **Wash ($0M): 30.0** cars/hr **(NEW BOTTLENECK!)**
- **Hand Dry ($3M):** $24.0 + 9.0(3) =$ **51.0** cars/hr

Why It Fails:

- Bottleneck shifts to **Wash (**30 **cars/hr)**!
- **21.0** cars/hr of Dry capacity sits **100% wasted**!
- **Profit:** $8 \times 30 = 240$ cars $\implies$ **$525/day** *(loses $300/day).*

FORMULA REMINDER: Vac $= 36 + 1.5(\$M)$ | Wash $= 30 + 15(\$M)$ | Dry $= 24 + 9(\$M)$ | Profit $= (\$5 \times \text{Cars}) - \$675$

Step 2B: What If We Split the $3M Evenly ($1M / $1M / $1M)?



Naive Try B: ($1.0M Vacuum, $1.0M Wash, $1.0M Dry)

- **Vacuum ($1M):** $36.0 + 1.5(1) = \mathbf{37.5}$ cars/hr
- **Wash ($1M):** $30.0 + 15.0(1) = \mathbf{45.0}$ cars/hr *(12/hr wasted!)*
- **Hand Dry ($1M):** $24.0 + 9.0(1) = \mathbf{33.0}$ cars/hr **(BOTTLENECK!)**

Profit: $8 \times 33 = 264$ cars $\implies$ **$645/day** *(loses $180/day).*

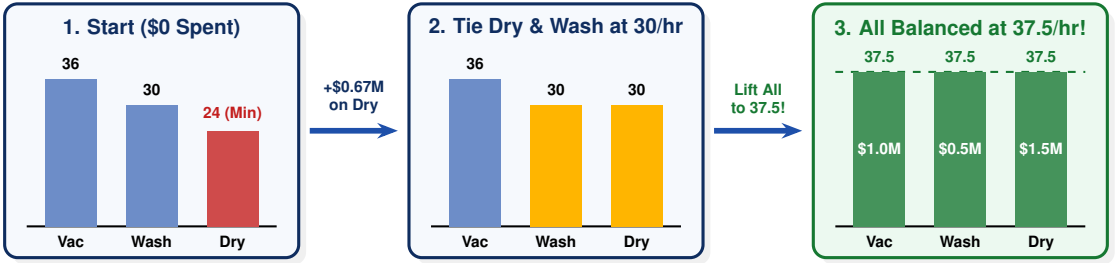
Golden Rule of Deterministic Capacity

Zero Capex is wasted ONLY when all 3 stages match:

Vacuum Speed = Wash Speed = Hand Dry Speed

FORMULA REMINDER: Vac $= 36 + 1.5(\$M)$ | Wash $= 30 + 15(\$M)$ | Dry $= 24 + 9(\$M)$ | Profit $= (\$5 \times \text{Cars}) - \$675$

Step 3: Spend Dollar by Dollar — Lifting the Lowest Bar!



Always Spend Your Next Dollar on the Lowest Bar:

1. **24 → 30/hr:** Spend \$0.67M on Dry $\implies$ **Dry & Wash tie at 30/hr**.
2. **30 → 36/hr:** Spend \$0.67M on Dry + \$0.40M on Wash $\implies$ **all 3 tie at 36/hr**!
3. **36 → 37.5/hr:** Remaining \$1.27M lifts all three together to **37.5** cars/hr!

FORMULA REMINDER: Vac $= 36 + 1.5(\$M)$ | Wash $= 30 + 15(\$M)$ | Dry $= 24 + 9(\$M)$ | Target Balanced Speed $= 37.5$ cars/hr

Step 4: The 60-Second Paper Exercise (Match All 3 to 37.5/hr)

Worked Example: Stage 1 (Vacuum)

- **Target Speed:** Customer demand is **37.5** cars/hr (300 cars/day).
- **Vacuum Equation:**

$$36.0 + 1.5 \times (\text{Vacuum \$M}) = 37.5$$

- **Solve for Vacuum $M:**

$$\text{Vacuum \$M} = \frac{37.5 - 36.0}{1.5} = \$\mathbf{1.00}\ \text{Million}$$

Your Turn (60 Seconds — 2 Quick Questions)

1. **Find Wash $M and Dry $M:** How much must you spend so Wash ($30 + 15 \times \$M = 37.5$) and Dry ($24 + 9 \times \$M = 37.5$) both hit 37.5 cars/hr?
2. **Find Max Daily Profit:** At $R = 37.5$ cars/hr (300 cars/day), what is your Daily Operating Profit?

FORMULA REMINDER: $\$M\ \text{Needed} = \frac{\text{Target Speed (37.5)} - \text{Base Speed}}{\text{Speed Gained per \$1M}}$ | $\text{Cars/Day} = 8 \times R$ | $\text{Profit} = (\$5 \times \text{Cars/Day}) - \$675$

Bonus Insight: Why Does Challenge 1 Start "Loaded" with 12 Cars?

What You See at 9:00 AM:

Challenge 1 starts with $\sim$ 12 **cars already inside** at 9:00 AM. Why?

1. Little's Law on Your Winning Screen

- **Throughput Rate (R):** 37.5 cars/hr
- **Cycle Time (CT):** $60/37.5 = \mathbf{1.6}$ min/car
- **Throughput Time (T): 19.0** min
- **Steady-State WIP:**

$$\text{WIP} = \frac{19.0 \text{ min}}{1.6 \text{ min/car}} = \mathbf{11.875} \approx \mathbf{12} \text{ cars!}$$

2. Avoiding Warm-Up Loss

- Starting empty (0 cars) means the 1st car wouldn't exit until **9:19 AM**!
- You would lose 12 cars warming up (288 cars vs. 300).
- Pre-loading **12** cars starts the day in **steady-state equilibrium**!

FORMULA REMINDER: Steady-State WIP $= \frac{\text{Throughput Time } (T{=}19.0 \text{ min})}{\text{Cycle Time } (CT{=}1.6 \text{ min/car})} = 11.875 \approx \mathbf{12}$ cars in system

Preparing for Challenge 2: Why a “Perfectly Balanced” Line Fails

Rated vs. Usable Capacity

- **Rated Capacity:** Max speed under zero variability (Challenge 1).
- **Usable Capacity:** Realistic average speed with random slowdowns (Challenge 2).
- With variability:

$$\text{Actual } R < \text{Usable Cap.} < \text{Rated Cap.}$$

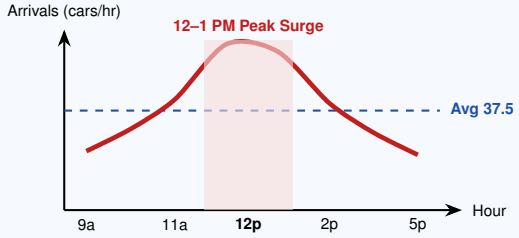
The 100% Utilization Trap

- Challenge 1 ran at **100% utilization** (37.5 cars/hr).
- In Challenge 2, **100% utilization causes gridlock**:
  - Slow hours immediately build a queue (WIP).
  - Zero spare sprint capacity to burn off the queue!

FORMULA REMINDER: Capacity Buffer $=$ Rated Capacity $-$ Average Demand | With Variability: Target Utilization (U) $< 100\%$

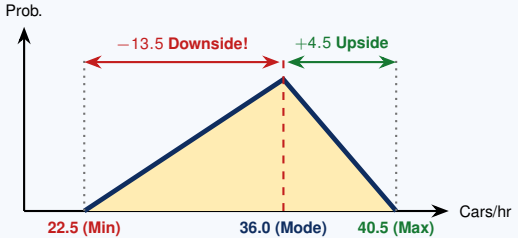
Anatomy of Variability in Challenge 2: Demand & Process

1. Demand Surge: 12–1 PM Rush



- Same $\sim$ 300 cars/day, but bunched at **11:30 AM – 1:30 PM**.

2. Process Skew: Heavy Downside Tail



- Vacuum ($0M): Upside $= +4.5$, but Downside $= -13.5$!

FORMULA REMINDER: Downside Risk = Mode − Min | Upside Gain = Max − Mode | When Downside > Upside: Average Speed < Mode Speed!

How Variability Destroys Throughput: Starving & Moving Bottlenecks



1. Upstream Starvation

When Vacuum slows down, Wash and Dry sit idle. **Unused bay time at 10 AM cannot be saved for 12 PM!**

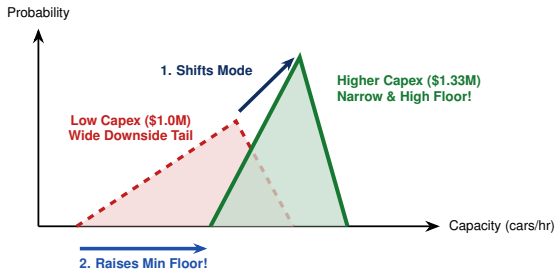
2. Midday Queue Spike

When Wash or Dry slows during the noon rush, WIP piles up and Wait Time (T) explodes.

3. Moving Bottleneck

Because station speeds fluctuate every hour, **the bottleneck wanders across stages**!

The Hidden Power of Capital in Challenge 2: Raising the Floor



Two Effects of Capex in Challenge 2

1. **Shifts Mode Right:** Higher peak speed.
2. **Raises the Minimum Floor:** Eliminates 22.5 cars/hr slowdowns!

Why Challenge 1's Split Failed

Spending only $1.0M on Vacuum left Stage 1 exposed to deep morning slowdowns—starving Wash and Dry!

Tactical Playbook for Graded Challenge 2

1. Protect Stage 1 (Anti-Starvation)

- **Vacuum is the gatekeeper:** Shift Capex **upstream into Vacuum (and Wash)** to raise their minimum floors.

2. Accept "Unbalanced" Modes

- Do *not* equalize Mode speeds! **Hand Dry** can run at a lower Mode if fed a steady stream of cars.

3. Check the "Compare" Tab Charts

- Watch the **12–1 PM peak** on the Hourly Utilization & WIP charts to see which stage chokes.

4. Test in $50k–$100k Steps

- Adjust budget in clean increments toward **270** cars/day (**$675** profit).

LAB SESSION 2 — Graded Challenge 2 (25 Minutes)

Your Graded Mission: Challenge 2 (Variation)

Allocate your \$3,000,000 budget in **Challenge 2 (GRADED)** using upstream floor protection and queue diagnostics.

Bronze Tier	Silver Tier	Gold Tier (HBP Optimum)
$\geq$ 255 **Cars/Day** ($\geq$ \$600 **Profit**)	$\geq$ 265 **Cars/Day** ($\geq$ \$650 **Profit**)	270 **Cars/Day** (\$675 **Profit**)
Beats Challenge 1's split.	Strong upstream protection.	Optimal variance buffering!

RECORD FOR GRADING: Best (Vacuum \$M, Wash \$M, Dry \$M) split, Cars Washed, Profit, CT, and T!

FORMULA REMINDER: (Vacuum \$M) + (Wash \$M) + (Dry \$M) = \$3.00M | Daily Profit = (\$5 $\times$ Cars Washed) − \$675

Final Debrief: Comparing Challenge 1 vs. Challenge 2 Strategies

1. Where Did Capex Move?

- Compare against Ch. 1's ($1.0M, $0.5M, $1.5M).
- Why did shifting money **from Dry to Vacuum/Wash** boost output?

2. "Flat Optimum" Effect

- Multiple nearby splits hit $\sim$ 270 cars ($675 profit).
- Moving bottlenecks create a **flat plateau** rather than a sharp peak.

3. Cost of Variability

- Same 300-car avg demand and $3M budget.
- Variability destroys **30 cars/day** and **$150/day** even under optimal management!

Core Lesson: Without variability, invest to **balance speeds**. With variability, invest to **raise upstream floors and buffer variance**.

Executive Summary: 4 Enduring Rules of Capacity Management

1. The Bottleneck Governs the P&L

- System capacity = **Lowest Stage Capacity**.
- Upgrading a bottleneck helps **only until the constraint shifts**; extra Capex beyond that is wasted.

2. Little's Law (WIP $= R \times T$) Is Ironclad

- $CT = 60/R$ is exit pace; T is total customer wait.
- Queue WIP directly inflates wait time ($T = \text{WIP} \times CT$).

3. Service Capacity Is Perishable

- Idle bay time at 10 AM cannot be stored for 12 PM.
- **Never starve downstream stages:** protect Stage 1 reliability.

4. Variability Requires Capacity Buffers

- Targeting **100% utilization** under variability guarantees gridlock.
- Invest to **shrink variance and raise the minimum floor**.